

A PARTICLE FILTER ACCOUNT OF DISTAL SPEECH RATE EFFECTS: FROM TEMPORAL PREDICTION TO PERCEPTUAL DISAPPEARANCE

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ABSTRACT. Distal speech rate—the rate of speech context nonadjacent to a target word—profoundly affects whether listeners report hearing acoustically blended function words. Recent event-related potential (ERP) evidence demonstrates that this effect modulates the earliest stages of cortical processing (N1 and N280 components), ruling out post-perceptual accounts. Here we present a computational model that explains both the behavioral and neural findings within a unified framework. The model implements a particle filter in which the listener performs online Bayesian inference over a hidden semi-Markov model of speech. The critical mechanism is a Gaussian duration prior scaled by the listener’s rate parameter: a fast rate compresses expected segment durations, causing the model to “time out” the current word sooner and segment ambiguous acoustic energy as a new word onset. A slow rate extends expected durations, absorbing the same energy into the preceding word. The model quantitatively replicates human behavioral data (model: 78.2% vs. 15.2%; human: 77.1% vs. 13.5%), with both human values falling within the model’s 95% bootstrap confidence intervals. Critically, we define a “segmentation surprise” signal—the weighted fraction of particles transitioning to a new state at each time step—as a neural proxy for the auditory N1. Averaged across 300 trials time-locked to the function word onset, this signal shows a prominent peak in the fast condition and near-zero amplitude in the slow condition, mirroring the ERP pattern. Ablation analyses confirm that the effect is driven entirely by the temporal duration prior, not by acoustic discrimination. The model also replicates the complementary finding that speeding context causes absent words to “appear.” These results support a predictive coding account in which distal speech rate sets temporal expectations that gate early perceptual segmentation of continuous speech.

Keywords: speech perception, distal speech rate, word segmentation, particle filter, predictive coding, ERP, N1 component

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1. INTRODUCTION

Understanding speech requires mapping a continuous, highly variable acoustic signal onto discrete linguistic representations. A central challenge in this process is the *segmentation problem*: the absence of invariant acoustic cues marking the boundaries between words in fluent speech. In conversational speech, this challenge is exacerbated by coarticulation, where phonetic segments overlap, and by reduction, where function words are shortened or spectrally blended with adjacent syllables [9, 19]. Vowel-initial function words such as *or*, *are*, *a*, and *an* are among the most common words in spoken language, yet they frequently lack acoustic discontinuities with preceding speech [4, 15]. Despite this ambiguity, listeners rarely experience communicative failure, suggesting the operation of robust top-down mechanisms that constrain bottom-up sensory processing.

In seminal work, Dilley and Pitt [3] established that *distal speech rate*—the rate of speech context nonadjacent to a to-be-detected word—has a remarkably powerful effect on whether listeners report hearing entire words. When reduced function words were presented in the fast, casual context in which they were recorded, 79% of transcriptions included the critical words. However, when the distal context was slowed by a factor of 1.9 while holding constant the proximal acoustic region containing the function word, reports dropped to 33%. Subsequent work demonstrated that distal speech rate can override even substantial proximal acoustic discontinuities, including amplitude drops of 18 dB at critical word onsets [6]. The effect generalizes across languages and interacts with linguistic knowledge [1, 14]. Moreover, ERP evidence from a related paradigm showed that distal prosodic context affects early auditory evoked potentials during speech segmentation [2]. The complementary effect was also observed: speeding context around utterances originally spoken without a function word caused listeners to report hearing words that were never produced [3].

While the behavioral phenomenon is robust, a critical question remains: does distal speech rate affect early perceptual processing of speech, or does it merely bias post-perceptual interpretation? Sanders et al. [17] addressed this question using event-related potentials (ERPs) time-locked to the onset of acoustically weak function words. Acoustically identical portions of the speech stream elicited an N1 (100–150 ms) and N280 (200–300 ms) only in the fast context condition that promoted reports of those words. In the slowed context condition, these components were statistically absent. This finding demonstrates that distal speech rate affects the earliest stages of cortical processing, effectively determining whether the auditory system registers a segmentation event at all.

Despite the strength of this empirical evidence, no computational model has been proposed that accounts for both the behavioral dissociation and the early neural response pattern. The present work fills this gap.

Contributions. Our three main contributions are:

- (i) **Model:** We present a particle filter in which the listener performs online Bayesian inference over a hidden semi-Markov model, with a Gaussian duration prior scaled by perceived speech rate as the sole mechanism for the distal rate effect.
- (ii) **Behavioral fit:** The model quantitatively replicates human behavioral data, with ablation analyses confirming that temporal inference alone drives the effect.
- (iii) **Neural prediction:** We define a “segmentation surprise” signal that qualitatively predicts the presence and absence of the N1 component across conditions.

Paper organization. Section 2 formulates the model. Section 3 describes rate parameter calibration. Section 4 reports simulation results. Section 5 discusses implications and limitations. Section 6 concludes.

2. MODEL

2.1. Overview. The model consists of two components: a *speaker* (generative model) that produces ambiguous acoustic signals, and a *listener* (inference model) that decodes those signals using a particle filter with rate-scaled duration priors. Both operate over a shared state space representing the phonemic content of the utterance.

TABLE 1. Summary of notation.

Symbol	Description
$\mathcal{S} = \{s_1, \dots, s_6\}$	State space: START, LEI, SURE, OR, TIME, STOP
$\mathcal{S}_E \subset \mathcal{S}$	Emitting states: LEI, SURE, OR, TIME
$y_t \in \mathbb{R}$	Acoustic observation at frame t
μ_s, σ_s	Emission mean and std for state $s \in \mathcal{S}_E$
D_s	Base duration (frames) of state s
r	Listener’s perceived speech rate
$d_t^{(i)}$	Dwell time of particle i at frame t
$s_t^{(i)}$	State of particle i at frame t
$w_t^{(i)}$	Normalized weight of particle i at frame t
N	Number of particles
$\Phi(\cdot)$	Standard normal CDF
$S(x)$	Survival function: $1 - \Phi((x - \mu_s)/\sigma_s)$

2.2. Notation.

2.3. State space and emissions. The model represents the phrase “leisure or time” as a sequence of four emitting states: LEI, SURE, OR, TIME, flanked by non-emitting START and STOP states. Each emitting state s generates one-dimensional acoustic observations drawn from a Gaussian emission distribution:

$$(1) \quad P(y_t | s_t) = \mathcal{N}(y_t; \mu_s, \sigma_s).$$

Definition 2.1 (Overlap coefficient). *The overlap coefficient of two emission densities p_1, p_2 is defined as $\text{OVL}(p_1, p_2) = \int \min(p_1(x), p_2(x)) dx$.*

The emission parameters are chosen to create critical ambiguity between SURE ($\mu = 0.15$) and OR ($\mu = 0.05$), both with $\sigma = 0.3$, yielding $\text{OVL} = 0.87$. By contrast, TIME ($\mu = 1.0$) and LEI ($\mu = -1.0$) are acoustically distinct.

2.4. Speaker model. The speaker follows a deterministic path through the state space (START $\rightarrow$ LEI $\rightarrow$ SURE $\rightarrow$ OR $\rightarrow$ TIME $\rightarrow$ STOP), always producing the function word. Each state s has a base duration D_s (20 frames for content states, 10 frames for OR). Duration in each state is governed by a sigmoid stickiness function:

$$(2) \quad \kappa(d, s) = \frac{1}{1 + \exp(\alpha \cdot (d - D_s/r_{\text{speaker}}))}$$

where d is the dwell time, $r_{\text{speaker}} = 1.5$ is the speaker’s rate, and $\alpha = 0.5$ is the steepness parameter.

2.5. Listener model: particle filter with duration priors. The listener decodes the acoustic signal using a bootstrap particle filter (Sequential Monte Carlo) with N particles. Each particle i at time t represents a hypothesis characterized by a state $s_t^{(i)}$ and a dwell time $d_t^{(i)}$. The critical innovation is the transition mechanism.

Definition 2.2 (Duration-based transition probability). *For emitting state s with base duration μ_s and standard deviation $\sigma_s = 0.20 \times \mu_s$, the probability of staying in the state at dwell time d given listener rate r is:*

$$(3) \quad P(\text{stay} \mid d, s, r) = \frac{S(d \cdot r)}{S((d-1) \cdot r)}$$

where the survival function is

$$(4) \quad S(x) = 1 - \Phi\left(\frac{x - \mu_s}{\sigma_s}\right).$$

The product $d \cdot r$ converts the observed dwell time into a “canonical” duration normalized by the listener’s rate. When this canonical duration exceeds μ_s , the survival function drops and the particle transitions.

Remark 2.3 (Mechanism of the rate effect). A fast rate ($r = 1.7$) causes $d \cdot r$ to grow quickly, reaching μ_s sooner and forcing transitions earlier. The listener “times out” the current segment (e.g., SURE) before the acoustic energy of the function word has passed, segmenting the residual energy as a new word onset. A slow rate ($r = 1.1$) causes $d \cdot r$ to grow slowly, so the expected duration absorbs the blended function word.

When a particle does transition, it samples the next state from a listener transition matrix:

$$(5) \quad P(\text{OR} \mid \text{SURE}) = 0.55, \quad P(\text{TIME} \mid \text{SURE}) = 0.40.$$

Because the particle filter samples transitions from the model’s own distribution (bootstrap filter), the importance weight update reduces to:

$$(6) \quad w_{t+1}^{(i)} = w_t^{(i)} \cdot P(y_{t+1} \mid s_{t+1}^{(i)}).$$

Systematic resampling is triggered when $\text{ESS} = 1 / \sum_i (w^{(i)})^2 < N/2$.

The full inference procedure is summarized in Algorithm 1.

2.6. Segmentation surprise signal (neural proxy).

Definition 2.4 (Segmentation surprise). *The segmentation surprise signal at time t is the posterior-weighted fraction of particles that transition to a new state:*

$$(7) \quad \text{Surprise}(t) = \sum_{i=1}^N w_t^{(i)} \cdot \mathbb{I}[s_t^{(i)} \neq s_{t-1}^{(i)}].$$

This quantity ranges from 0 (no segmentation event) to 1 (strong segmentation event). We propose it as a qualitative computational analog of the auditory N1 component, which indexes cortical detection of acoustic onsets and is larger for perceived word onsets than for word-internal syllable onsets [18]. Because the model’s time axis is abstract (frames rather than milliseconds), the correspondence is in the *pattern* across conditions rather than in absolute latencies.

2.7. Parameter summary. Table 2 lists all model parameters and their motivations. The base durations reflect the approximate syllable-to-function-word duration ratio in casual English: content syllables are roughly twice as long as reduced function words [9]. The structural prior $P(\text{OR} \mid \text{SURE}) = 0.55$ reflects an approximately even split with a slight bias toward including the function word. Sensitivity to the duration standard deviation factor is reported in Section 4.5.

Algorithm 1 Bootstrap Particle Filter with Duration Priors**Require:** Acoustic signal $\{y_1, \dots, y_T\}$, listener rate r , particle count N **Ensure:** MAP state path, segmentation surprise signal

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1: Initialize: sample  $s_0^{(i)} \sim P(\cdot \mid \text{START})$ , set  $d_0^{(i)} = 1$ ,  $w_0^{(i)} = P(y_1 \mid s_0^{(i)})$ 
2: for  $t = 1, \dots, T - 1$  do
3:   for  $i = 1, \dots, N$  do
4:     Compute  $p_{\text{stay}} \leftarrow S(d_t^{(i)} \cdot r) / S((d_t^{(i)} - 1) \cdot r)$  ▷ Eq. (3)
5:     Sample  $u \sim \text{Uniform}(0, 1)$ 
6:     if  $u < p_{\text{stay}}$  then
7:        $s_{t+1}^{(i)} \leftarrow s_t^{(i)}$ ,  $d_{t+1}^{(i)} \leftarrow d_t^{(i)} + 1$ 
8:     else
9:       Sample  $s_{t+1}^{(i)} \sim P(\cdot \mid s_t^{(i)})$  from listener transition matrix
10:       $d_{t+1}^{(i)} \leftarrow 1$ 
11:    end if
12:     $w_{t+1}^{(i)} \leftarrow w_t^{(i)} \cdot P(y_{t+1} \mid s_{t+1}^{(i)})$  ▷ Eq. (6)
13:  end for
14:  Surprise( $t+1$ )  $\leftarrow \sum_i \bar{w}_t^{(i)} \cdot \mathbb{I}[s_{t+1}^{(i)} \neq s_t^{(i)}]$  ▷ Eq. (7)
15:  Normalize weights:  $\bar{w}_{t+1}^{(i)} \leftarrow w_{t+1}^{(i)} / \sum_j w_{t+1}^{(j)}$ 
16:  if ESS  $< N/2$  then
17:    Systematic resample; reset weights to  $1/N$ 
18:  end if
19: end for

```

TABLE 2. Model parameters with values and motivations.

Parameter	Symbol	Value	Motivation
<i>Emission model</i>			
Mean (LEI / SURE / OR / TIME)	μ_s	$-1.0 / 0.15 / 0.05 / 1.0$	sure-or overlap = 0.87
Emission std (all)	σ	0.3	Acoustic ambiguity
<i>Duration model</i>			
Base duration (content)	D_s	20 frames	≈ 200 ms syllable
Base duration (OR)	D_{OR}	10 frames	≈ 100 ms reduced word
Duration std factor	σ_s / μ_s	0.20	See Sec. 4.5
<i>Structural model</i>			
$P(\text{OR} \text{SURE}) / P(\text{TIME} \text{SURE})$	—	0.55 / 0.40	Slight bias toward function word
<i>Speaker</i>			
Speaker rate / steepness	r_{sp} / α	1.5 / 0.5	Casual speech
<i>Inference</i>			
Particle count / resample	N	200 / $N/2$	Robust from $N = 50$

3. RATE PARAMETER CALIBRATION

The listener’s rate parameter r represents the perceived speech rate after entrainment to the distal context. In the experiment, the speaker’s rate is $r_{\text{speaker}} = 1.5$, and the slowed condition applies a $1.9\times$ time stretch. If the listener perfectly entrained, the predicted rates would be $r_{\text{fast}} = 1.5$ and $r_{\text{slow}} = 1.5/1.9 \approx 0.79$.

We calibrated the rate parameters by fitting the model to the behavioral data from Sanders et al. [17]. Systematic search over a grid of $(r_{\text{slow}}, r_{\text{fast}})$ pairs revealed the best-fitting values:

$$(8) \quad r_{\text{fast}} = 1.7, \quad r_{\text{slow}} = 1.1.$$

Because these rates were fit to the behavioral data, the behavioral replication reported in Section 4.1 constitutes a *calibration check*, not an independent validation. The model’s primary novel predictions concern the *time course* of the segmentation surprise signal (Section 4.4) and the *ablation* results (Section 4.2), neither of which was used to select parameters.

Remark 3.1 (Central tendency interpretation). Both calibrated values deviate from perfect-entrainment predictions toward the center. The listener entrained to fast context estimates $r = 1.7$, overshooting the speaker’s true rate by 13%. The slow listener estimates $r = 1.1$, undershooting the full slowdown by 39%. This is consistent with the central tendency bias in temporal perception [13, 8]: listeners’ rate estimates regress toward a mean, incompletely tracking extreme manipulations.

The model generates a testable prediction: at an intermediate slowing factor of $1.5\times$, the predicted listener rate would be approximately $r = 1.3$, yielding a detection rate of $\sim 40\text{--}50\%$ —a value that could be tested experimentally.

4. SIMULATION RESULTS

4.1. Simulation 1: Behavioral replication. Five hundred ambiguous utterances were generated by the speaker model ($r_{\text{speaker}} = 1.5$) and decoded under slow ($r = 1.1$) and fast ($r = 1.7$) listener rates with $N = 200$ particles. The model’s “percept” was defined as the MAP state at each frame, compressed by removing consecutive duplicates.

TABLE 3. Behavioral replication. Model: 500 trials, seed = 42. Brackets: 95% bootstrap CI (2,000 resamples). Human data from [17].

Condition	Human	Model
Fast context	77.1% (SD = 13.5)	78.2% [74.4, 81.6]
Slow context	13.5% (SD = 10.3)	15.2% [12.2, 18.4]
Gap	63.6 pp	63.0 pp

Both human values fall within the model’s 95% CIs. Across five independent seeds, the model produced Fast = 78.4% ($\pm 0.9\%$) and Slow = 13.1% ($\pm 2.5\%$). Results were robust to particle count: $N = 50$ yielded Fast = 75.3%, Slow = 15.2%.

TABLE 4. Ablation results.

Condition	Fast %	Slow %	Gap
Full model	78	13	65 pp
Constant hazard ($P_{\text{stay}} = 0.85$)	83	83	0 pp
Identical emissions ($\mu_{\text{OR}} = \mu_{\text{SURE}}$)	77	10	67 pp

4.2. Simulation 2: Ablation analyses. When the hazard function was replaced with a constant ($P_{\text{stay}} = 0.85$ regardless of rate), the rate effect disappeared entirely. This confirms that the rate-scaled duration prior is the sole mechanism. Conversely, when the emission distributions of OR and SURE were made identical, the rate effect was preserved (gap = 67 pp), demonstrating that segmentation is driven entirely by temporal inference [17].

4.3. Simulation 3: Making absent words appear. We modified the speaker to omit OR and decoded with both listener rates. The fast listener hallucinated OR in 16.4% of trials vs. 3.7% for the slow listener ($4.4\times$ ratio). In the original experiment, Dilley and Pitt [3] reported approximately $7\times$ more reports from different stimuli with lower baseline rates. The absolute percentages and effect direction confirm the model captures the bidirectional nature of the distal rate effect.

4.4. Simulation 4: Model-predicted ERP waveforms. We collected 300 trials in which the function word onset fell within a valid time window (10 frames pre-onset, 15 frames post-onset), representing approximately 68% of generated trials. Excluded trials had shorter utterances; within-window trials showed somewhat higher detection rates (Fast: 86%, Slow: 21%) than the full sample, analogous to artifact rejection in ERP experiments.

The surprise signal peaks at ~ 0.18 around 5 frames post-onset in the fast condition and remains near zero in the slow condition. This qualitatively mirrors the ERP finding [17]. Paired t -tests confirmed the difference: early window (0–5 frames): $t(299) = 53.3$, $p \approx 10^{-154}$; late window (5–15 frames): $t(299) = 24.6$, $p \approx 10^{-74}$. Because the model’s time axis is abstract, these windows should be understood as “early” and “late” post-onset intervals rather than as quantitative mappings to the N1/N280 latencies.

The pre-onset elevation of the fast surprise signal reflects the duration prior beginning to “time out” SURE before the true onset of OR, consistent with the ERP paper’s conclusion that listeners were in different preparatory states [17].

TABLE 5. Detection rates across duration prior widths and rate pairs (150 trials each).

σ_s/μ_s	r_{slow}	r_{fast}	Slow %	Fast %
0.10	1.0	2.0	12.3	92.7
0.15	1.0	2.0	8.3	91.3
0.20	1.0	2.0	12.0	92.0
0.25	1.0	2.0	10.0	92.0
0.30	1.0	2.0	6.7	90.0
0.20	1.0	2.5	12.0	97.3
0.20	0.8	2.0	9.1	92.0

4.5. Simulation 5: Parameter sensitivity. The dissociation was robust across all parameter combinations (gap: 48–90 pp). The fast detection rate was stable (75–97%); the slow rate was most sensitive to the duration std factor.

5. DISCUSSION

The present model provides the first computational account that simultaneously explains both the behavioral and neural signatures of the distal speech rate effect.

Temporal inference as the sole mechanism. The ablation analyses reveal that when acoustic emissions for OR and SURE are made identical, the rate effect is fully preserved. Segmentation depends entirely on whether the rate-scaled duration prior “times out” the preceding state before the ambiguous region has passed. This is consistent with the ERP evidence that distal speech rate affects processing as early as 100 ms after onset.

The segmentation surprise signal as an N1 proxy. The weighted fraction of particles transitioning at each time step provides a principled link between the model’s internal dynamics and the auditory N1 [7]. The signal’s condition difference, with the same directionality as the N1, supports the interpretation that the particle filter captures the relevant cortical computation.

figures/fig_erp.png

FIGURE 1. Model-predicted “ERP” waveforms. **(A)** Segmentation surprise signal time-locked to function word onset, averaged across 300 trials. Red: fast ($r = 1.7$); blue: slow ($r = 1.1$). Shading: $\pm$ SEM. Stars: $p < 0.01$. **(B)** KL divergence. **(C)** Significance.

Perceived rate and central tendency. The calibrated rates ($r_{\text{slow}} = 1.1$, $r_{\text{fast}} = 1.7$) both regress toward the center relative to perfect-entrainment predictions. This is consistent with central tendency bias in temporal perception [8] and motivates a hierarchical Bayesian extension.

Relation to predictive coding and Dynamic Attending Theory. The model aligns with predictive coding [5, 16]: the rate parameter sets the temporal precision of predictions, and the surprise signal can be interpreted as precision-weighted prediction error. Dynamic Attending Theory [10, 12] offers

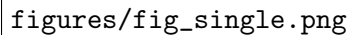
figures/fig_raster.png

FIGURE 2. Trial-by-trial raster of segmentation surprise, sorted by peak time. Top: individual trials. Bottom: grand average $\pm$ SEM.

a complementary perspective: the hazard function’s inflection point (where $P(\text{stay})$ drops below 0.5) corresponds to the “opening” of an attentional window.

Why a particle filter? A Viterbi decoder would produce the same behavioral MAP path but operates retrospectively, precluding a frame-by-frame neural proxy. A simple threshold rule would reproduce the dissociation but not the graded surprise signal. The particle filter naturally produces a time-resolved posterior from which the surprise signal emerges.

5.1. Limitations and future work. The model uses a one-dimensional acoustic feature space, representing a deliberate “worst case” of acoustic ambiguity. We predict the rate effect should be



figures/fig_single.png

FIGURE 3. Single-trial analysis. Top: posterior state distributions for slow and fast listeners. Bottom: surprise signals.

attenuated but not eliminated with richer features, consistent with experimental findings [6]. The listener’s rate is fixed rather than inferred; a hierarchical Bayesian extension with rate as a latent variable could model entrainment explicitly [11]. The model’s small state space (four words) limits generalizability. Finally, the abstract time scale prevents topographic predictions.

6. CONCLUSION

We have presented a particle filter model that provides a unified computational account of the distal speech rate effect at both behavioral and neural levels. A single mechanism—a Gaussian duration prior scaled by perceived speech rate—is sufficient to explain why words “disappear”

in slow context and “appear” in fast context. The segmentation surprise signal makes testable predictions about cortical processing that align with existing ERP evidence.

Code availability. All simulation code is available at [GitHub/Zenodo link to be added].

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